

Zhuoyuan Xu

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Educational Background

Shanghai Jiao Tong University, Undergraduate

Sept 2022 - Jun 2026

- *Major:* Computer Science. (*John Hopcroft Honors Class, top 5%* SJTU students in computer science. The bottom 20% of the students are eventually forced to leave the class.)
- *Core courses:* Linear Algebra(97), Machine Learning Theory(96), Algorithm(94), Introduction to Theoretical Computer Science(95), Modern Algebra(93), Information Theory(95)
- Conducted research with Professor Yu Yu.

National University of Singapore, Research Assistant

Aug 2025 - Feb 2026

- Conducted research with Professor Jiaheng Zhang.

Research Experience

LigeSIS: Distribution-friendly Polynomial Commitment Based on Error-correcting

Aug 2025 - Jan 2026

Code (Advisor: Prof. Jiaheng Zhang, NUS)

- *Field:* Applied cryptography. (Submitted paper for USE'NIX 26)
- *Content:* Proposed LigeSIS, the first error-correcting-code-based polynomial commitment scheme designed to be distributed-system friendly; developed a preprocessing-based acceleration that *reduces hashing overhead by 8×*; substantial improvements in *proof-generation time and communication cost*.
- *Contributions:* **Core author**. Co-designed a **Rust-based** distributed polynomial commitment algorithm based on Subset-Sum Hash, solving the challenge that traditional Merkle trees cannot be algebraically aggregated in distributed environments.

Proof for the Soundness of First-Order Logic (Advisor: Prof. Qinxiang Cao)

Dec 2023 - Jan 2024

- *Field:* Programming language, specifically formal verification. (Theory based, *intense mathematics*)
- *Content:* Proved that if a conclusion can be derived from premises via a formal proof system, then it holds in every model satisfying those premises (if $\Phi \vdash \phi$, then $\Phi \models \phi$).
- *Contributions:* Constructed abstract syntax *from scratch* to optimize variable binding algorithms; rigorously derived complex substitution lemmas; implemented a recursive deduction system in Coq; successfully proved the Soundness Theorem via strict structural induction.

Project Experience

Summer Individual Project: Scheme Compiler

Jun 2023 - Jul 2023

- *Advanced Engineering Techniques:* Architected a robust multi-pass compiler pipeline that decomposes complex translations into small, verifiable steps. Solved x86-64 hardware constraints (e.g. immediate size limits, two-operand arithmetic) via a custom instruction selection pass and implemented a closure conversion phase to transform high-level functions into flat code blocks with explicit environment capture, *ensuring modularity and testability across 40+ passes*.
- *Algorithm Implementation:* Designed and implemented *a graph-coloring register allocator* based on optimistic coloring. Performed live analysis to construct conflict graphs and handling high-degree nodes via spilling to stack frames, utilizing an iterative fixpoint algorithm to resolve conflicts between spillable and unspillable variables.
- *Fast-learning Capabilities:* Recognized that *standard conditional logic was error-prone* for complex Abstract Syntax Tree transformations; quickly mastered and integrated Scheme’s match library to replace verbose if/cond structures, significantly reducing code complexity and improved the robustness of the instruction selection and verifier passes.

Freshman Individual project: Minivim

Oct 2022 - Dec 2022

- *Engineering Techniques:* Integrated *core formatting operations* such as insertion, deletion, line breaks, and alignment using a mode-switching idea.

Extracurricular Activities

The 2024 Cryptography Summer Camp, Qizhi Institution

July 2024

Three-School "Growing Partners" Summer School, SJTU, THU, and PKU

July 2023

Honors and Awards

Zhiyuan Honors Scholarship (Top 10%), SJTU, 4 years in a row.

2022-2026

Shanghai Jiao Tong Contract Bridge Competition, Rank 3/14.

May 2025

University Texas Hold'em Competition (QRT), Rank 6/104.

Apr 2025

National Contract Bridge Tournament - EW Group, Rank 1 in SJTU, Top 1% nationwide

May 2024

China National Mathematics Olympics Beijing Area, Second Prize

Sep 2020

Personal Skills

Programming Languages: C++(skilled), Python(skilled), LaTeX(skilled), Rust, Coq, Scheme

English Proficiency: TOEFL(104)